MILITARY SPECIFICATION

RANGEFINDER SYSTEM, FIRE CONTROL: LASER AN/GVS-3

This amendment forms a part of Military Specification MIL-R-50608(MU) dated 22 September 1970.

Paragraph 3.5: Environmental.- Add the following:

"NOTE: In Configuration I as defined in 6.3.9, the system shall be mounted to a surface using the mounting hardware designated when being subjected to shock and vibration. For the remaining environmental tests of Configuration I, no mounting surface is required and only the chest with contents, with the chest cover closed, are to be exposed."

Paragraph 3.5.6: Delete and substitute the following:

"3.5.6 Humidity.- The system in Configuration A through H shall be capable of withstanding 10 cycles of exposure to the following:

a. Two hours of increasing temperature to +105°F and 95 percent relative humidity.

b. Sixteen hours at 105°F and 90 percent relative humidity.

c. Two hours decreasing temperature from 105°F to 70°F and increasing relative humidity to 95 percent.

d. Four hours at 70°F and 95 percent relative humidity."

3.5.6.1 Subsequent to exposure to humidity, the system shall not have been damaged or adversely affected and in Configurations A through E shall meet the applicable requirements of 3.5.12 through 3.10.3 within 30 minutes after exposure."

Paragraph 3.5.9.1 Operating.- Delete entire paragraph.

Paragraph 3.5.10 Fungus.- Delete and substitute the following:

"3.5.10 Fungus.- Items 1 through 16, excluding items 10 and 11, shall be capable of withstanding 28 days exposure to a composite spore suspension consisting of the following cultures: (a) Chaetomium globosum, (b) Aspergillus versicalor, (c) Aspergillus flavus, (d) Penicillium luteum, (e) Rhizopus nigricans. The composite spore suspension shall be incubated at an ambient temperature of between +77°F and +84°F with an RH of 97 ± 2%. Subsequent to the above fungus exposure, the items

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shall not have been damaged or adversely affected and when assembled into Configurations A through E, shall meet the requirements of 3.5.12 through 3.10.3."

Paragraph 3.7.1.1 Ranging.- Delete all after heading.

Paragraph 3.7.1.1.1 Ranging accuracy.- Delete "20,000 meters" and substitute "5,000 meters".

Add sub-paragraph "(3)" to 3.9c, as follows:

"(3) The Mount Adapter 10541111 shall be capable of interfacing with and securing to the Mount Shelf Assembly 10541103."

Paragraph 4.3.2.1: Delete all after 2nd sentence and add new paragraph 4.3.2.2 as follows:

"4.3.2.2 Rangefinder system.- Examination and tests related to Section 3 herein, shall be performed in accordance with the Quality Assurance Provisions Tables I to V inclusive in Section 4. Except where otherwise specified or necessarily applicable for environmental testing, examination and tests on production rangefinder systems shall be accomplished with Configuration A (see 3.4 and 6.3.1) to determine compliance with 3.7."

NOTE: When rangefinder systems are procured in accordance with this specification, the following test methods and procedures in Specification MIL-T-60500 shall not be required for inspection of the Transceiver 10540501 when provided as a part of the rangefinder system:

Test Methods & Procedures (MIL-T-60500A)	Paragraph
a. Target discrimination	4.6.5.3.2
b. Minimum range selection	4.6.5.3.3
c. Rapid ranging	4.6.5.3.4
d. Delay ranging	4.6.5.3.5
e. Illumination	4.6.5.5.1
f. Elevation accuracy	4.6.5.5.2
g. Elevation backlash	4.6.5.5.3
h. Elevation knob torque	4.6.5.5.4
i. Lateral displacement	4.6.5.5.5
j. Open sight	4.6.5.6.1
k. Plumb travel	4.6.5.6.4
l. Eyepiece focus	4.6.5.6.5
m. Diopter adjustment	4.6.5.6.6
n. Reticle illumination	4.6.5.6.7
o. Safety operation	4.6.5.9.1
p. Eye filter	4.6.5.9.2
q. Operational temperatures (-65° &+155°F)	4.6.2.9
r. Storage temperature (-80° & +160°F)	4.6.2.1
s. Humidity	4.6.2.4
t. Sand and dust	4.6.2.5
u. Salt spray	4.6.2.6
v. Rain	4.6.2.8
w. Fungus	4.6.2.12
x. Electromagnetic compatibility	4.6.8

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Table I "Classification of Defects": Delete Characteristics 101 "Ranging", 138 "Weight", 140 "Interchangeability" and 144 "Operational Temperatures"

Add Note at bottom of Table I as follows:

"NOTE: The tests of Table I shall be conducted at a temperature between +70 & 80 F."

Paragraph 4.4.1.2: Delete in its entirety (including Table III).

Table IV: Delete in its entirety and substitute the following:

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
305.	Interchangeability	3.9	4.6.9
306.	Storage temperatures (-80° and +160°F)	3.5.11	4.6.2.1
307.	Operating temperatures (-25°F, -65°F, +155°F)	3.5.12	4.6.2.13
308.	Vibration (-65° ± 2°F)	3.5.1	4.6.2.3
309.	Rain	3.5.4	4.6.2.5
310.	Submersion	3.5.5	4.6.2.6

Add paragraph 4.4.2.1 as follows:

"4.4.2.1 - Three rangefinder systems shall be selected at random as a special sampling at 25%, 50% and 75% completion of the contract production. Each sample shall be subjected to the tests in Table V and shall then meet the requirements and tests in Table I.

TABLE V

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
311.	Shock (between 70°& 80°F)	3.5.2	4.6.2.2
312.	Rough handling	3.5.3	4.6.2.4
313.	Humidity	3.5.6	4.6.2.7
314.	Sand and dust	3.5.7	4.6.2.8
315.	Salt spray	3.5.8	4.6.2.9
316.	Altitude (non-operating)	3.5.9.2	4.6.2.11

Add paragraph 4.5.2.3.3 as follows:

"4.5.2.3.3 Mount Shelf Assembly.- Two test devices shall be designed to simulate the maximum and minimum tolerance conditions of Mount Shelf Assembly, 10541103 as defined in 4.6.9.5.1 to determine interface with Mount Adapter, 10541111 in conformance with 3.9c(3)".

Paragraph 4.6.2: delete and substitute:

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"4.6.2 Environmental.- With the exception of Reliability testing (see 4.6.3) wherein all requirements of 3.7 through 3.10.3 must be met after each individual environmental test under 3.5 as a part of First Article Testing, all performance testing in accordance with 4.6.4 through 4.6.10.2 shall be conducted once after completion of all tests in 4.6.2."

Paragraph 4.6.2.3: Delete "Method 510" and substitute "Method 514".

Paragraph 4.6.2.7 Humidity.- Delete the sentence and substitute the following:

"The humidity test shall be performed as outlined in MIL-STD-810, Procedure V and under the conditions specified in 3.5.6."

Paragraph 4.6.2.8, Add: During the sand and dust test all exposed optical elements shall be protected with a masked covering.

Paragraph 4.6.2.10 Altitude-operating.- Delete entire paragraph.

Paragraph 4.6.2.12: Delete and substitute the following:

"4.6.2.12 Fungus.- One of each item listed in 1 through 16 (see 3.4) excluding items 10 and 11, shall be subjected to the fungus test outlined in MIL-F-13926 as part of the First Article test only. At the end of the 28 day incubation period, each item shall be visually examined for evidence of fungi and shall then be assembled into the configurations specified in 3.5.10 and subjected to the tests specified in 4.6.4 through 4.6.10.2 (less fungus) for compliance with 3.5.10. Subsequent acceptance of the system shall be by certification."

Paragraph 4.6.3, last line: Delete "(paragraph 3)" and substitute "(3.6)".

Paragraph 4.6.4.1 Ranging.- Delete all after heading.

Paragraph 4.6.4.2 Ranging accuracy.- In the first sentence delete targets at "5340, and 6450 meters" and substitute "4540, and 5000 meters". Also, in line 4 (5000 in lieu of 6450).

Paragraph 4.6.7: Add, "For the purposes of First Article and Reliability testing, test results of individual sample batteries shall be accumulated during the performance tests subsequent to each environmental exposure (see 4.6.2). Each sample rangefinder shall be fired with its battery until the completion of the performance checks after exposure to a particular environment. When a low voltage indication is indicated by the rangefinder, the number of rangings performed by the particular battery, as well as other information required under Reliability testing, will be recorded and the battery recharged for subsequent environmental exposure."

Paragraph 4.6.9.1, lines 1 and 2: Delete and substitute,"Three items shall be selected as follows from the systems to be presented as the first article sample (see 4.2) to determine compliance with 3.9 and subsequent inspection in accordance with Table IV in Section 4."

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Paragraph 4.6.9.1a: Delete last line and substitute,"selection as follows and the contractor shall then assemble the 3 groups into 3 rangefinder systems for inspection and testing in conformance with 4.6.9.2.

Paragraph 4.6.9.2, delete and substitute:

"4.6.9.2 System.- The 3 systems obtained from 4.6.9.1 shall be subjected to all requirements and tests of this specification in compliance with 3.9. Subsequent to selection (4.6.9.1) the conduct of final assembly, examination and tests shall be witnessed by the Quality Assurance Representative. After Government approval and acceptance of the first article (4.2), interchangeability tests shall be conducted in accordance with sampling in 4.4.2. When the manufacturing process, materials, or inspection techniques previously applied are subject to change, or an adverse change occurs in performance, capability or reliability of the system, this shall be cause to cease those operations contributing to the change or failure until corrections have been made by the contractor and approved by the Government."

Add paragraph 4.6.9.5.1 as follows:

"4.6.9.5.1 Mount shelf assembly.- Proper interfacing between the Mount Adapter, 10541111 and Mount Shelf Assembly, 10541103 shall be determined by utilizing test devices representing the Mount Shelf Assembly. Two (2) test devices shall be designed; one within 10% of the maximum limit and one within 10% of the minimum limit of tolerancing of the pertinent interfaces as determined from drawings F10541103 (Mount Shelf Assembly), F10541104 (Base), B10541091 (Plug), and B10541101 (Stud). When assembled, the cam lock lever of the adapter mount shall be capable of securing and releasing with the shelf mount. The contact surfaces, including the two (2) projecting studs in the shelf mount shall permit ready assembly without interference and all three (3) locating pads shall be in positive contact with their related interfaces on the plate of the adapter mount."

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